PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

SarthiX-RaahSathi

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"SarthiX-RaahSathi"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Shikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura | **Ashish Yadav (2210991404)**
Date: ..**28-04-2026**.. | **Hitesh Saini(2210991670)**

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

The project titled **"Sarthix"** aims to design and develop a comprehensive web-based logistics marketplace platform that streamlines and automates transportation and shipment management operations between drivers and shippers. The system provides a centralized platform for shipment posting, real-time bidding, permit compliance monitoring, trip management, and driver performance tracking.

The primary objective of the project is to eliminate dependency on informal logistics networks, improve pricing transparency, enhance operational efficiency, and ensure regulatory compliance through digital transformation. The platform enables shippers to post shipments and compare bids from multiple drivers, while drivers can discover available loads, place competitive bids, and track their earnings and trip history.

The application incorporates secure authentication mechanisms, role-based access control, and an intuitive user interface to ensure smooth interaction for both drivers and shippers. It also includes advanced features such as a transparent bidding system, shipment lifecycle management, driver ratings, notification systems, and automated permit expiry alerts to help drivers avoid penalties and legal risks.

The system has been developed using modern web technologies including React, Clerk authentication, and Supabase PostgreSQL, ensuring scalability, security, and high performance. The implementation demonstrates how digital marketplace solutions can significantly improve efficiency, transparency, and compliance within the logistics and transportation industry.

1. Introduction

1.1. Overview of the Project

Sarthix – A Smart Logistics Marketplace and Permit Compliance System is a web-based application designed to modernize and streamline logistics and transportation operations between drivers and shippers. In the traditional logistics industry, shipment discovery, pricing negotiation, and permit compliance are often managed through informal networks or disconnected systems, leading to inefficiencies, inconsistent earnings for drivers, lack of pricing transparency, and increased operational risks.

The proposed system aims to provide a comprehensive digital platform that integrates all essential logistics operations into a single unified marketplace. By leveraging modern web technologies, the application enables seamless interaction between drivers and shippers through a transparent bidding-based shipment management system. It ensures that shipment details, bids, permits, trips, and notifications are stored in a structured and easily accessible format, improving coordination and reducing manual dependency.

The system centralizes key functionalities such as shipment posting, driver bidding, permit management, trip tracking, and performance analytics. Drivers can browse available shipments, place competitive bids, and monitor their earnings and completed trips, while shippers can compare bids, assign drivers, and track shipment progress efficiently. In addition, the platform introduces an automated permit expiry notification system that alerts drivers before permits expire, helping them maintain regulatory compliance and avoid penalties.

The application also incorporates secure authentication mechanisms, role-based access control, and an intuitive user interface to ensure ease of use for both drivers and companies. Developed using modern technologies such as React, Clerk Authentication, and Supabase PostgreSQL, the system ensures scalability, security, and high performance.

Overall, Sarthix contributes to the digital transformation of the logistics industry by improving operational efficiency, increasing pricing transparency, enhancing driver management, and reducing legal risks associated with expired permits.

1.2. Problem Statement

Despite advancements in technology, the logistics and transportation industry still heavily relies on informal communication channels and fragmented systems for shipment management and driver coordination. These traditional approaches often create challenges such as lack of pricing transparency, inefficient shipment discovery, delayed communication, and difficulty in tracking operations effectively.

Drivers frequently depend on brokers or personal networks to find shipment opportunities, resulting in inconsistent earnings and limited access to reliable work. On the other hand, shippers face difficulties in comparing multiple drivers, evaluating bids transparently, and selecting the most suitable transporter for their shipments. This lack of a centralized system leads to inefficiencies, delays, and reduced operational visibility.

Another major concern in the logistics sector is permit compliance management. Vehicle permits and transportation documents are often tracked manually, making it difficult for drivers and companies to monitor expiry dates effectively. Expired permits can result in heavy fines, legal penalties, operational disruptions, and delays in shipment delivery. Existing logistics platforms mainly focus on load matching and often lack integrated compliance monitoring systems.

Additionally, many current systems do not provide proper role-based access control, centralized data management, or real-time notifications, which can lead to data inconsistency and poor coordination between drivers and shippers.

Therefore, there is a strong need for a scalable, secure, and intelligent logistics management platform that can address these challenges efficiently. The system should provide transparent shipment bidding, centralized shipment and driver management, automated permit expiry notifications, real-time operational updates, and a seamless user experience for both drivers and shippers.

1.3. Objectives of the Project

The primary objective of **Sarthix** is to develop an efficient and reliable logistics marketplace platform that simplifies and automates various transportation and shipment management processes. The system is designed with the following specific objectives:

● To create a centralized platform for managing logistics-related activities, including shipment posting, driver bidding, trip management, and permit compliance monitoring

● To reduce dependency on informal logistics networks and minimize operational inefficiencies caused by manual coordination and fragmented communication

● To implement secure authentication and role-based access control using modern technologies such as Clerk Authentication to ensure secure access and data protection

● To enable a transparent real-time bidding system where drivers can compete fairly for shipments and shippers can compare bids efficiently

● To provide automated permit expiry notifications that help drivers maintain regulatory compliance and avoid legal penalties or operational delays

● To offer shipment lifecycle tracking and performance analytics, allowing shippers to monitor deliveries and drivers to track earnings, completed trips, and ratings

● To design a user-friendly and responsive interface that improves usability and accessibility for both drivers and shippers across different devices

● To ensure scalability of the platform so that additional features such as GPS tracking, payment integration, AI-based pricing, and real-time communication can be incorporated in the future

These objectives collectively aim to improve operational efficiency, enhance transparency in logistics operations, increase driver income visibility, and provide a more organized and reliable transportation management system for both drivers and shippers.

1.4. Scope of the Project

The scope of **Sarthix** includes the design and development of a fully functional logistics marketplace platform that addresses the core requirements of shipment management, driver coordination, and regulatory compliance within the transportation industry. The system primarily focuses on connecting drivers and shippers through a transparent bidding mechanism while enabling efficient shipment tracking and permit management.

The application allows shippers to register, log in, create shipment requests, manage shipment details, compare driver bids, and assign shipments to suitable drivers. Drivers can browse available shipments, place bids, manage vehicle permits, track completed trips, and monitor their earnings and ratings. The system also provides automated permit expiry notifications to help drivers maintain compliance and avoid legal or operational issues.

Administrators and system stakeholders can oversee platform activities, monitor operational data, and ensure smooth workflow management and data consistency. The centralized architecture improves communication, transparency, and operational efficiency between all participants in the logistics ecosystem.

While the current implementation focuses on essential logistics functionalities, the system is designed with scalability and extensibility in mind. Future enhancements may include advanced features such as GPS-based live shipment tracking, real-time chat between drivers and shippers, online payment gateway integration, AI-based pricing recommendations, fraud detection mechanisms, and driver verification systems.

The project is currently implemented as a web-based platform; however, it can be extended into mobile applications to improve accessibility and user engagement for drivers and companies on the move. Furthermore, integration with third-party logistics services, mapping APIs, and government permit verification systems can significantly enhance the platform’s capabilities.

In summary, the scope of this project is to provide a strong technological foundation for a modern logistics management and transportation marketplace system that can evolve with future technological advancements and increasing industry demands.

2. Methodology

2.1. Development Approach

The development of **Sarthix** follows the **Agile software development methodology**, which emphasizes iterative development, flexibility, collaboration, and continuous improvement throughout the project lifecycle. Unlike traditional software development models such as the Waterfall approach, Agile enables the system to evolve gradually through multiple development cycles, allowing the team to adapt to changing requirements and improve features based on continuous feedback.

The project was divided into smaller functional modules, and each module was designed, developed, tested, and refined incrementally. Major modules included user authentication, shipment management, bidding functionality, permit compliance monitoring, notification systems, and dashboard analytics. Regular team discussions and iterative testing helped identify issues early, improve system performance, and ensure smooth integration between components.

The Agile approach also supported efficient collaboration and task distribution among team members, enabling parallel development of different parts of the application such as frontend development using React, backend and database integration using Supabase PostgreSQL, and authentication management using Clerk. This iterative workflow improved productivity, accelerated problem-solving, and ensured the delivery of a scalable and high-quality logistics marketplace platform.

Additionally, Agile methodology allowed continuous enhancement of user experience by incorporating feedback during development, ensuring that the platform remained aligned with the practical needs of both drivers and shippers.

2.2. Requirement Analysis

Requirement analysis is an important phase in the development process where the needs of drivers, shippers, and administrators are identified and documented. The requirements were analyzed by studying logistics workflows and shipment management processes.

The requirements were divided into two categories:

2.2.1 Functional Requirements

- User registration and authentication
- Shipment posting and bidding system
- Driver and permit management
- Shipment tracking and notifications
- Dashboard for managing operations and analytics

2.2.2 Non-Functional Requirements

- Security and data protection
- Scalability and reliability
- User-friendly interface
- Fast and efficient data management

This phase ensured that the system was designed to effectively address real-world logistics challenges while providing a secure, scalable, and efficient platform for both drivers and shippers.

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2.3. System Design

The system design phase involved creating the architecture and structure of the application to ensure scalability, efficiency, and maintainability.

2.3.1 Architecture Design

The application follows a client-server architecture:

- The frontend handles user interaction and dashboard management
- The backend manages business logic and request processing
- The database stores shipment, bidding, user, and permit data

2.3.2 Database Design

A relational database schema was designed using PostgreSQL (Supabase) to efficiently manage users, shipments, bids, trips, permits, and notifications. Proper relationships and constraints were implemented to maintain data integrity and reduce redundancy.

2.3.3 User Interface Design

The user interface was designed to be responsive and user-friendly for both drivers and shippers. Wireframes and layouts were planned before development to ensure consistency and smooth navigation across the platform.

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2.4. Development Process

The development phase involved implementing the designed system using modern web technologies.

2.4.1 Frontend Development

The frontend was developed using React with Vite, providing a fast and interactive user interface. Responsive design techniques were used to ensure accessibility across different devices.

2.4.2 Backend Development

The backend functionality and database operations were managed using Supabase, which handled server-side logic, APIs, and real-time data management.

2.4.3 Database Integration

PostgreSQL (via Supabase) was used as the database to efficiently manage users, shipments, bids, permits, trips, and notifications while maintaining data consistency and integrity.

2.4.4 Authentication and Authorization

Clerk Authentication was implemented to provide secure login, user management, and role-based access control for drivers and shippers.

The collaborative development approach ensured smooth integration of all modules and cohesive system functionality.

2.5. Testing and Deployment

Testing is an essential phase to ensure the reliability and performance of the system. Various testing techniques were used to identify and resolve issues.

2.5.1 Testing

● Unit Testing: Individual modules such as bidding, permit management, and authentication were tested for correct functionality

● Integration Testing: Interaction between frontend, backend, authentication, and database services was verified

● User Testing: The platform was tested from driver and shipper perspectives to ensure usability and smooth workflow

Bugs and errors identified during testing were resolved to improve system stability, security, and performance.

2.5.2 Deployment

After successful testing, the application was deployed using cloud-based platforms:

● The frontend, developed using React with Vite, was deployed on platforms such as Vercel for fast performance and scalability

● Backend services, authentication, and database management were handled using Supabase and Clerk, providing secure and scalable infrastructure

Deployment ensured that the platform was accessible online and capable of handling real-world logistics operations efficiently.

3. Tools and Technologies

The frontend of **Sarthix** is developed using React with Vite. React provides a component-based architecture for building dynamic and interactive user interfaces, while Vite ensures fast development and optimized performance. The frontend is designed to deliver a smooth and responsive experience for both drivers and shippers.

Responsive UI design techniques are used to ensure accessibility across different screen sizes and devices. Together, these technologies help create a modern, user-friendly, and efficient logistics platform.

3.2 Backend

The backend services are managed using Supabase, which provides scalable backend functionality including database management, APIs, and real-time data handling. Supabase simplifies development by offering integrated backend services and reducing server-side complexity.

Clerk Authentication is integrated to handle secure user authentication, session management, and role-based access control for drivers and shippers.

3.3 Database

PostgreSQL is used as the primary relational database for storing structured data such as user information, shipments, bids, permits, trips, ratings, and notifications. It ensures data integrity, reliability, and efficient query execution.

Supabase, built on top of PostgreSQL, is also utilized as the database platform, providing additional features such as real-time updates, automatic API generation, and seamless integration with the application.

3.1. Security

Security is an important aspect of the system, and **Clerk Authentication** is used for secure authentication and authorization. It ensures that only authorized users can access specific features based on their roles, such as driver or shipper.

Additionally, Supabase security mechanisms help protect sensitive data, manage user sessions securely, and enforce database-level access control.

3.4 Additional Tools and Services

Several additional tools and services are used to enhance the functionality and deployment of the system:

● **Supabase Realtime** is used for real-time updates such as bid changes, shipment updates, and notifications.

● **Vercel** is used for deploying the frontend application, providing fast performance and scalable hosting.

● **Git and GitHub** are used for version control and collaboration among team members, enabling efficient code management and teamwork.

These tools collectively improve development efficiency, platform performance, scalability, and overall system reliability.

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4. Implementation

4.1. Architecture

The **Sarthix** system is designed using a client-server architecture, which ensures a clear separation between the user interface and backend services. This architecture improves scalability, maintainability, and overall system performance.

In this model, the frontend (client-side), developed using React with Vite, handles user interaction and dashboard functionality for drivers and shippers. It communicates with the backend services through APIs and real-time database operations.

The backend infrastructure is managed using Supabase, which handles business logic, database communication, authentication, and real-time updates efficiently. Operations such as user authentication, shipment posting, bidding, permit management, and notification handling are processed securely through backend services.

API communication plays a vital role in enabling smooth data exchange between the frontend and backend. Features such as shipment bidding, permit alerts, trip tracking, and dashboard analytics rely on secure and efficient data synchronization.

PostgreSQL, integrated through Supabase, is used for structured data storage, ensuring data consistency, integrity, and reliability. The overall architecture is modular and scalable, allowing future enhancements such as GPS tracking, payment integration, AI-based pricing, and real-time communication features to be added easily.

4.2. Features

The **Sarthix** system incorporates a comprehensive set of advanced features designed to improve logistics operations, automate workflows, and ensure secure and efficient shipment management:

● Secure authentication and role-based access control using Clerk Authentication, providing separate access permissions for drivers and shippers to ensure data security and controlled system access.

● A transparent real-time bidding system that allows drivers to compete for shipments while enabling shippers to compare bids and select suitable drivers efficiently.

● A centralized shipment management workflow that handles the complete shipment lifecycle, including shipment posting, bidding, driver assignment, trip completion, and status tracking.

● A PostgreSQL-based relational database architecture that ensures data consistency and integrity across users, shipments, bids, permits, trips, ratings, and notifications.

● Automated permit compliance monitoring system that tracks permit validity and generates expiry alerts to help drivers avoid penalties and operational delays.

● Real-time notification system for bid updates, shipment status changes, and permit expiry alerts, improving communication between drivers and shippers.

● Driver performance analytics dashboard that provides insights into completed trips, earnings, active bids, and ratings for better performance tracking.

● Responsive and user-friendly interface designed for seamless access across multiple devices and screen sizes.

● Cloud-based deployment and scalable infrastructure using Vercel, Supabase, and Clerk to ensure reliability, fast performance, and efficient data handling.

● CI/CD integration using GitHub Actions for automated testing and deployment, ensuring smoother updates and consistent application releases.

● Shipment image management system that allows shippers to upload shipment-related images for better load visibility and transparency.

● Role-based workflow management that dynamically controls features and dashboard access according to user type and operational requirements.

Custom prescription builder that allows users to create and manage personalized prescription templates for flexible and standardized clinical documentation.

5. Results

5.1. Efficiency

The implementation of **Sarthix** has significantly improved the efficiency of logistics and transportation operations. By automating processes such as shipment posting, bidding, permit monitoring, and trip management, the system reduces dependency on manual coordination and informal communication methods. This improves workflow efficiency, reduces delays, and minimizes operational errors.

Drivers can easily discover shipment opportunities and track their earnings, while shippers can efficiently compare bids, assign drivers, and monitor shipment progress. The streamlined workflow ensures better utilization of resources and faster decision-making.

5.2 Data Management

The system provides a centralized database for storing and managing logistics-related data, including user information, shipments, bids, permits, trips, ratings, and notifications. This centralized structure eliminates data redundancy and ensures consistency across the platform.

Users can access and update information in real time, improving coordination between drivers and shippers. PostgreSQL, integrated through Supabase, ensures reliable data storage and efficient retrieval of operational information whenever required.

5.3 Security

Security is a critical aspect of the system, and Sarthix ensures strong protection of user and operational data through secure authentication mechanisms. Clerk Authentication and Supabase security features ensure that only authorized users can access the platform.

Role-based access control restricts functionalities according to user roles such as driver or shipper, preventing unauthorized access and improving data privacy and integrity. Overall, the system provides a secure and reliable environment for managing logistics operations and sensitive business data.

Future Scope

The **Sarthix** system provides a strong foundation for digital logistics and transportation management; however, there are several opportunities for future enhancement and expansion to make the platform more advanced and comprehensive.

● Integration of GPS-based live tracking for real-time shipment monitoring and route optimization.

● Real-time chat and communication modules between drivers and shippers for faster coordination and issue resolution.

● AI-based dynamic pricing and bidding recommendations to improve pricing transparency and operational efficiency.

● Payment gateway integration for secure online payments, driver settlements, and transaction management.

● Advanced analytics and predictive insights for shipment trends, driver performance, and demand forecasting.

● Enhanced fraud detection and driver verification mechanisms to improve platform trust and security.

● Mobile application development for Android and iOS to improve accessibility and user engagement.

● Integration with government permit verification systems and third-party logistics APIs for automated compliance management.

● Automated route optimization and fuel efficiency analysis to reduce operational costs and improve delivery performance.

These enhancements can transform Sarthix into a fully intelligent, scalable, and technology-driven logistics marketplace platform.

6. Conclusion

The **Sarthix – Smart Logistics Marketplace and Permit Compliance System** successfully demonstrates how digital technologies can transform traditional logistics and transportation management practices. The system effectively addresses the limitations of manual coordination and fragmented logistics processes by providing a centralized, secure, and efficient platform for managing shipments, bidding, permits, and driver operations.

Through the implementation of modern web technologies such as React, Supabase, PostgreSQL, and Clerk Authentication, the project achieves scalability, reliability, and high performance. Key functionalities such as transparent shipment bidding, permit expiry notifications, shipment lifecycle management, driver analytics, and secure authentication significantly improve operational efficiency and user experience for both drivers and shippers.

The system reduces dependency on informal logistics networks, minimizes operational delays, improves pricing transparency, and enhances coordination between stakeholders. Automated permit monitoring further helps drivers maintain regulatory compliance and avoid legal or operational risks.

This project reflects a structured and collaborative development approach, where different modules such as frontend development, backend integration, database management, authentication, and deployment were implemented cohesively to ensure smooth system functionality.

In conclusion, Sarthix provides a practical, scalable, and technology-driven solution for modern logistics management and serves as a strong foundation for future advancements in the transportation and logistics industry.

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- React Documentation. Available at: https://react.dev

8. Appendices

Appendix A: System Screenshots

Figure A.1: Login Page

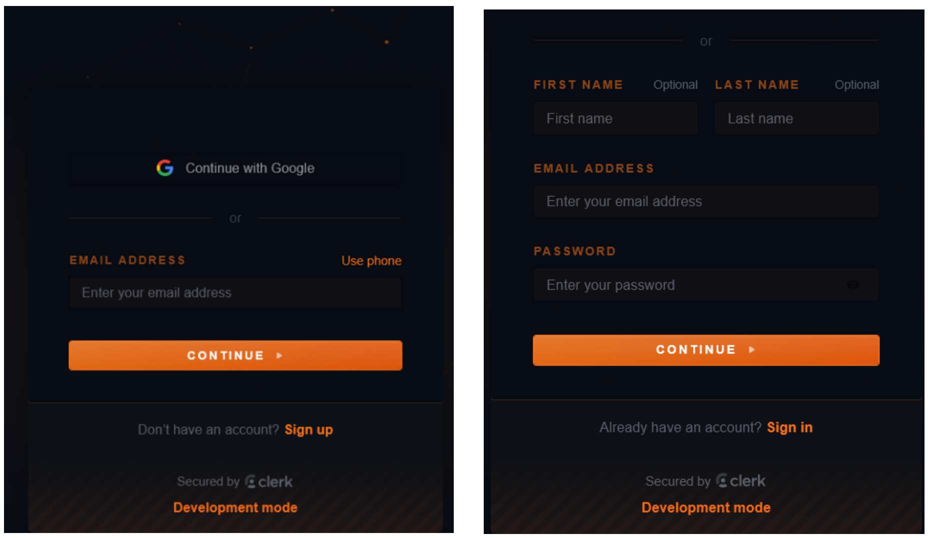


Figure A.2: Landing Page

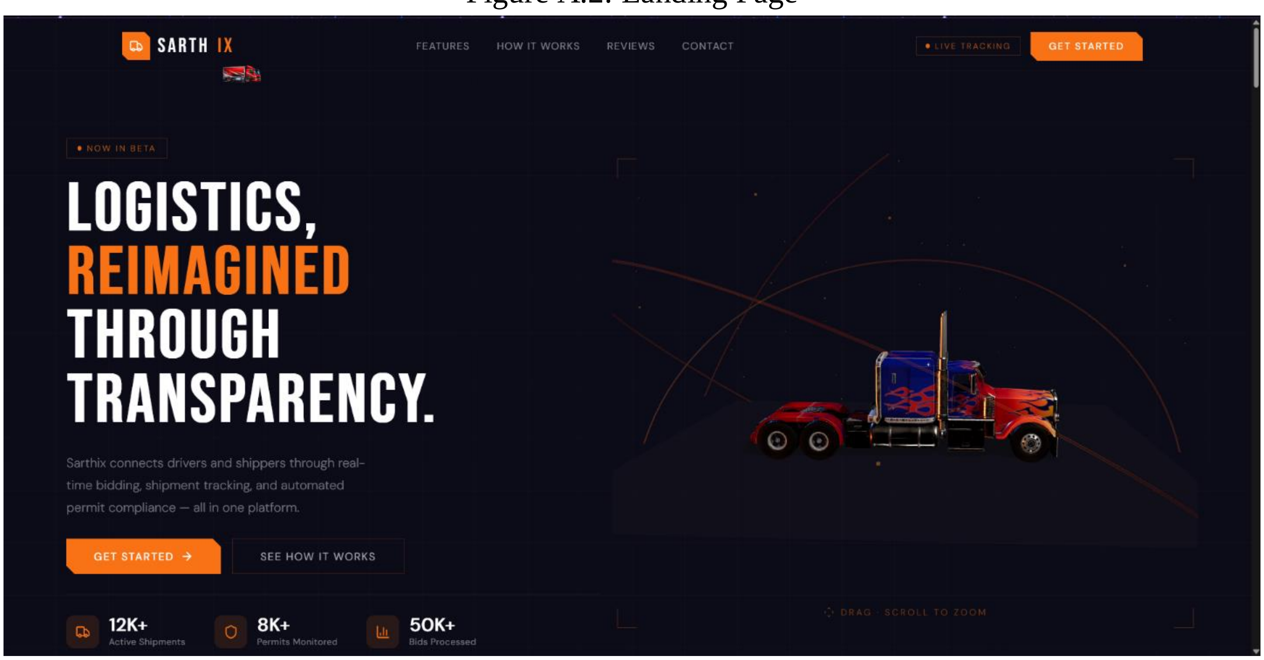


Figure A.3: Driver Dashboard

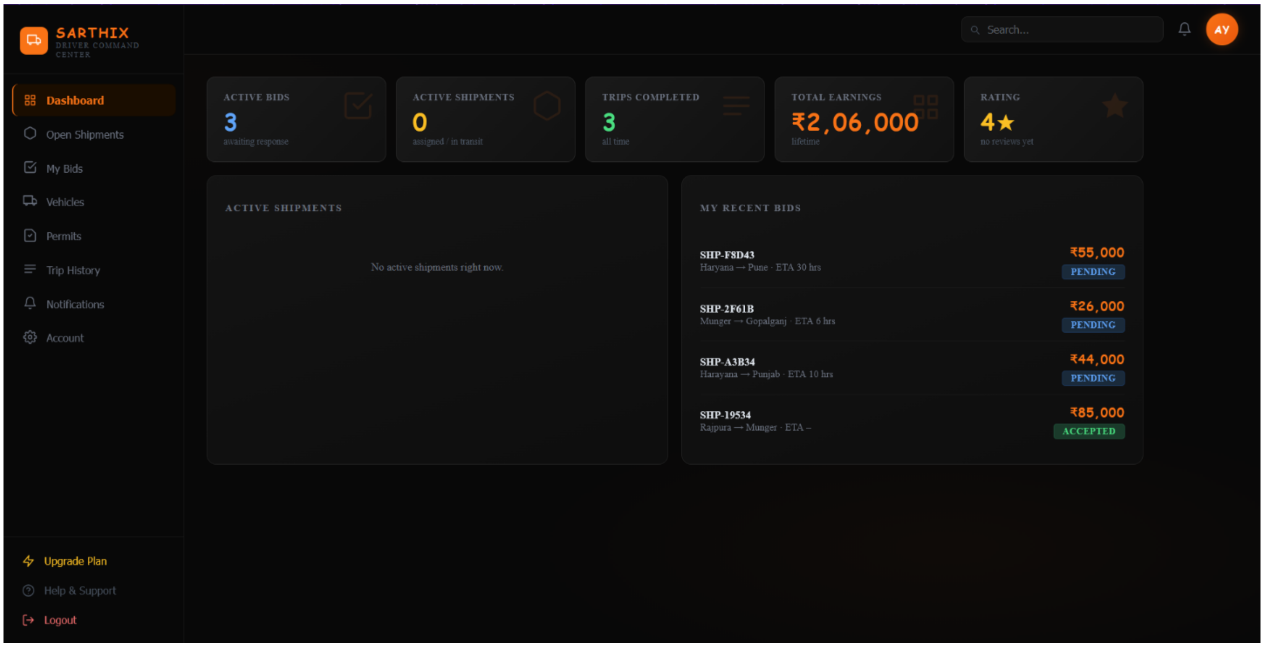


Figure A.4: Shipper Dashboard

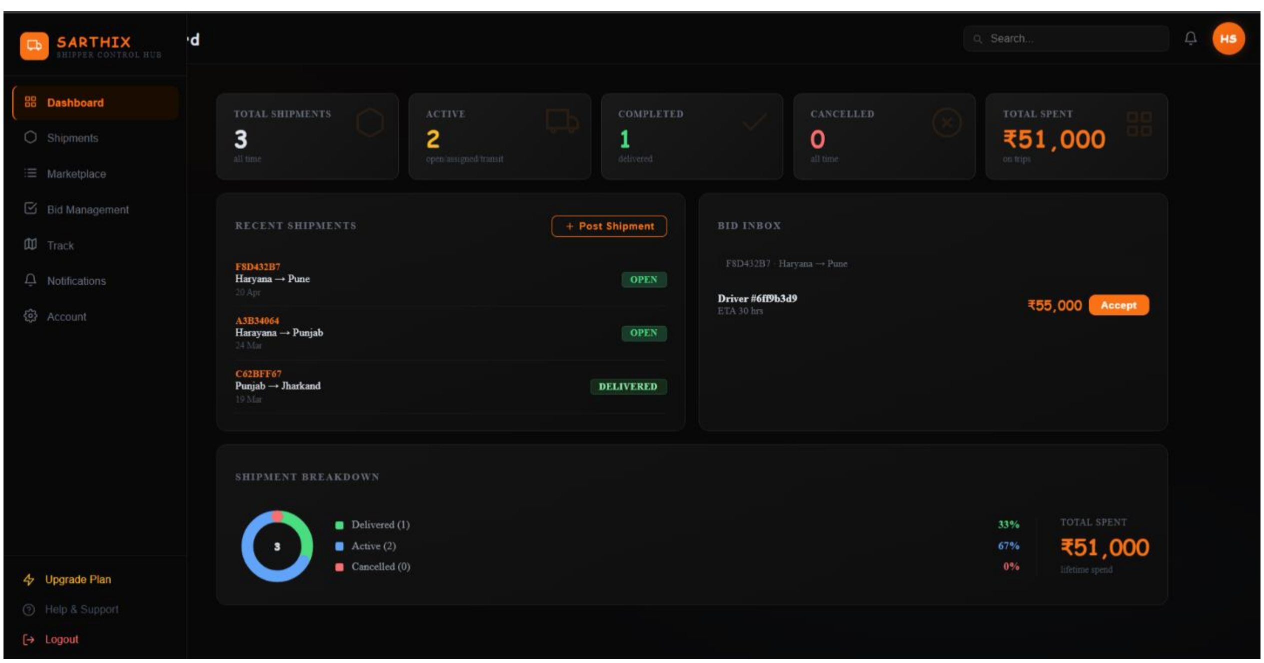


Figure A.5: Permit Section

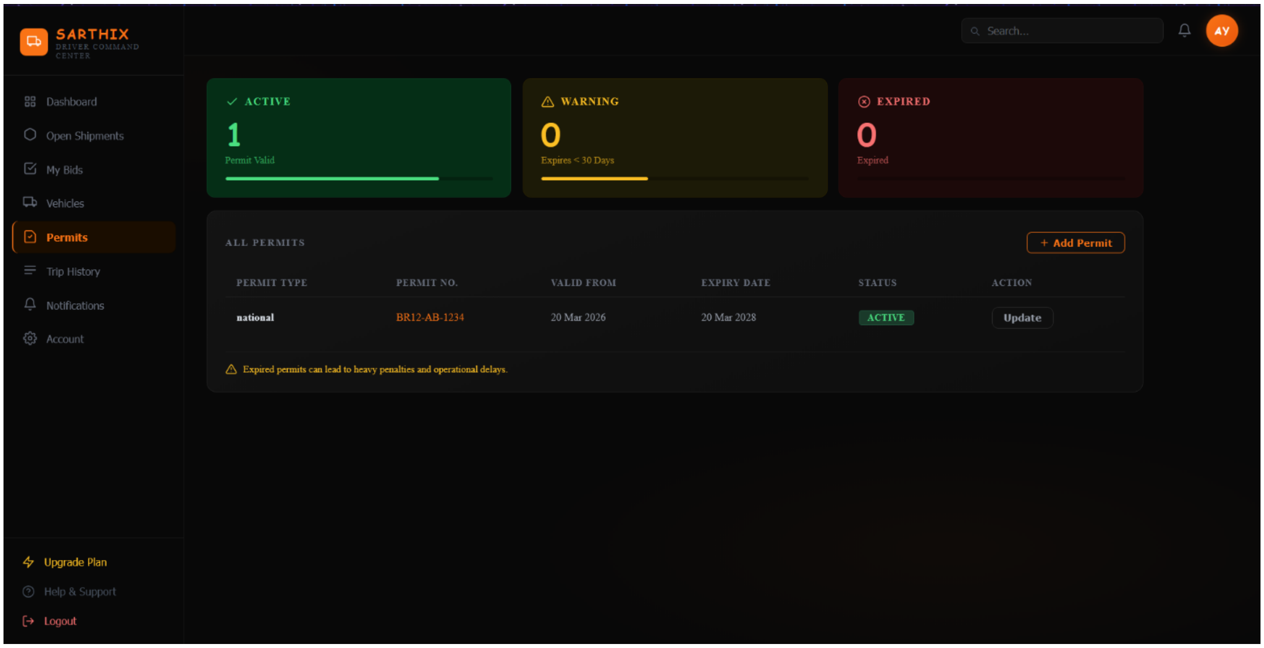
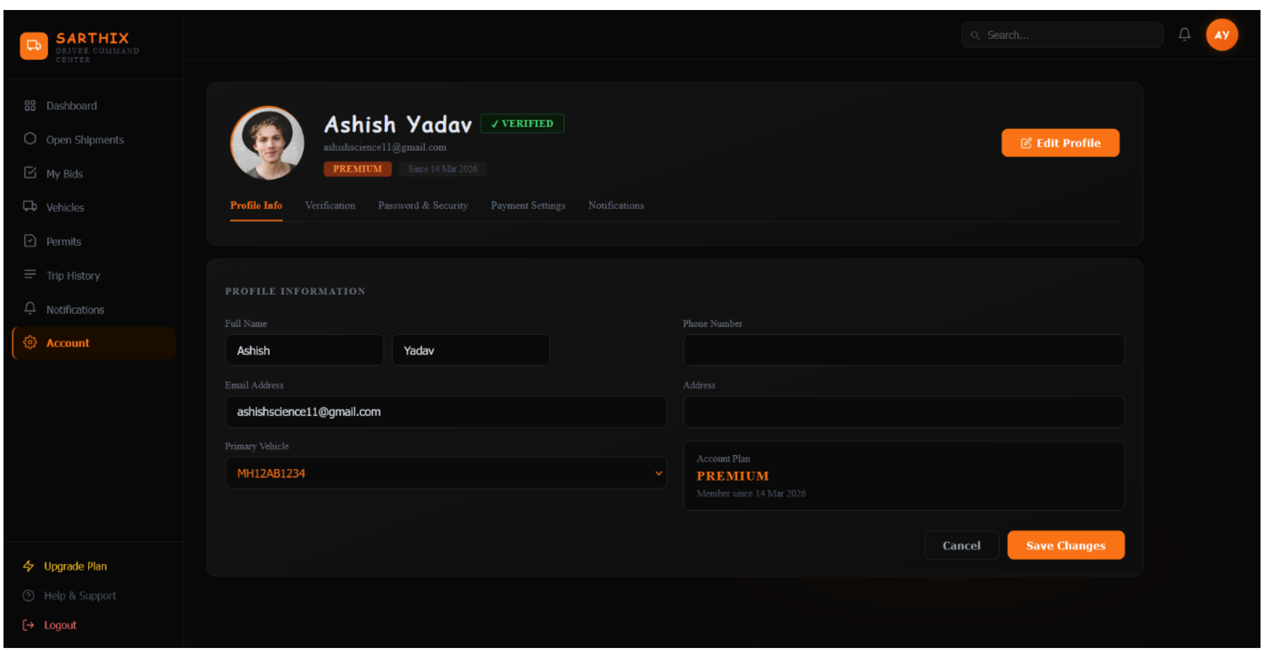


Figure A.5: Account



Appendix B: Database Schema

Figure C.1: ER Diagram